

USL – upper limit

LSL – lower limit

Mean value:

$$\bar{x} = \frac{1}{n} \sum_{i=0}^n x_i$$

Overall standard deviation:

$$\sigma_o = \sqrt{\frac{1}{n} \sum_{i=0}^n (x_i - \bar{x})^2}$$

Within standard deviation:

$$\sigma_w = \sqrt{\frac{1}{n-1} \sum_{i=0}^n (x_i - \bar{x})^2}$$

PP and PPK:

$$PP = \frac{USL - LSL}{6 \cdot \sigma_o}$$

$$PPL = \frac{\bar{x} - LSL}{3 \cdot \sigma_o}$$

$$PPU = \frac{USL - \bar{x}}{3 \cdot \sigma_o}$$

$$PPK = \min(PPL, PPU)$$

CP and CPK:

$$CP = \frac{USL - LSL}{6 \cdot \sigma_w}$$

$$CPL = \frac{\bar{x} - LSL}{3 \cdot \sigma_w}$$

$$CPU = \frac{USL - \bar{x}}{3 \cdot \sigma_w}$$

$$CPK = \min(CPL, CPU)$$

Stability:

$$\bar{s} = \frac{\max \text{ value}}{\min \text{ value}} - 1$$